

How I webscraped starting blocks:

Starting Blocks is a website created by the Australian Children's Education and Care Quality Authority (ACECQA) to provide transparency to child care pricing and quality information to decrease the information asymmetry in the child care market. It contains a list of all child care centres who are supported by the CCS (Child Care Subsidy). Centres are required to update their prices once every 6 months or so. At its peak, 6000 centres are being recorded out of 9,600 CCS-registered centre-based day care. It suggests that the below scrape may be incomplete, and difficult to clean.

I chose to scrape price data since the closes child care pricing data publicly available is from the Department of Education which only provides data at a Statistical Area 3 level, which is too large. A t-test on scraped vs DoE data confirmed my scrape was positively biased by a 44.6 cents. I include some correspondence in the appendix below with more information on this scrape and daily fee calculations with the StartingBlocks and Department of Education.

The scrape is divided into two steps: 1) identify the web ID for all centres, 2) scraping each individual webpage.

SECTION 1: identifying all websites codes

Step 1: I found there was a pattern to every html linked to the service ID of each centre. Using Algolia I could search API keys.

Step 2: I build a longitude and latitude search grid in a box of Australia and force run the code across every 0.4 degrees which searches within a 25km radius for centres.

Step 3: Every successful search records the service ID, public ID, provider ID, name, rating, rating number and coordinates so I can match back to the ACECQA quality dataset. Importantly, the public ID is what is used to locate the webpage of every child care centre.

This script is located in phase1_historical_fees. It took a few hours to run and would crash so I embedded an autosave function. Notably, the level of searching Claude coded seems like it would have been too specific and computationally-heavy so improvements can be made here.

Step 4: In phase2_historical_fees, it builds each individual website by:
<https://startingblocks.gov.au/find-child-care/{publicId}>

SECTION 2: scraping the individual centre

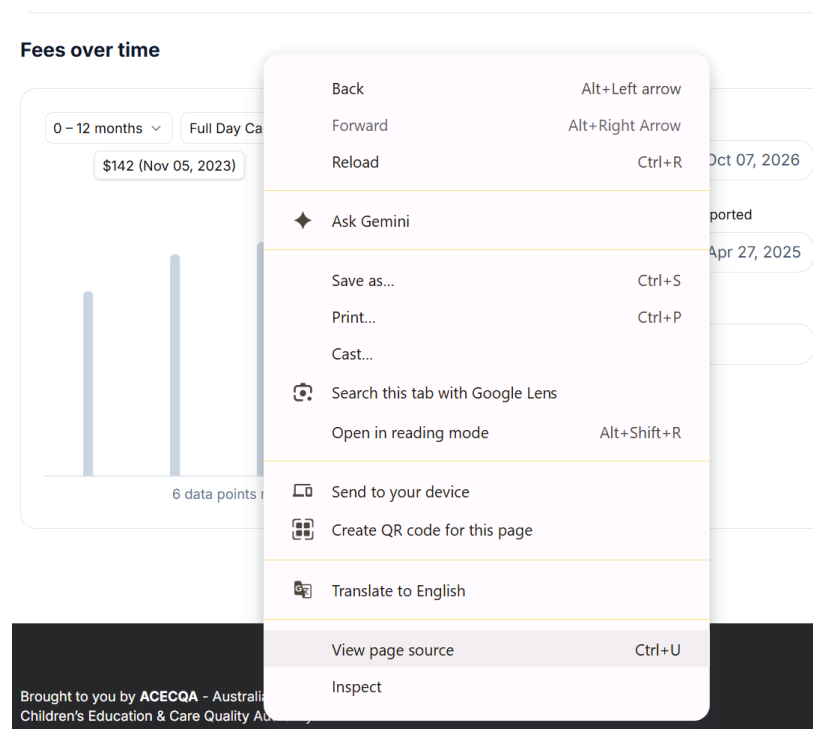
Step 1: Open Startingblock.com.au and select a random centre. I work with an example below... scrolling to the bottom of the page I observe:

Fees over time

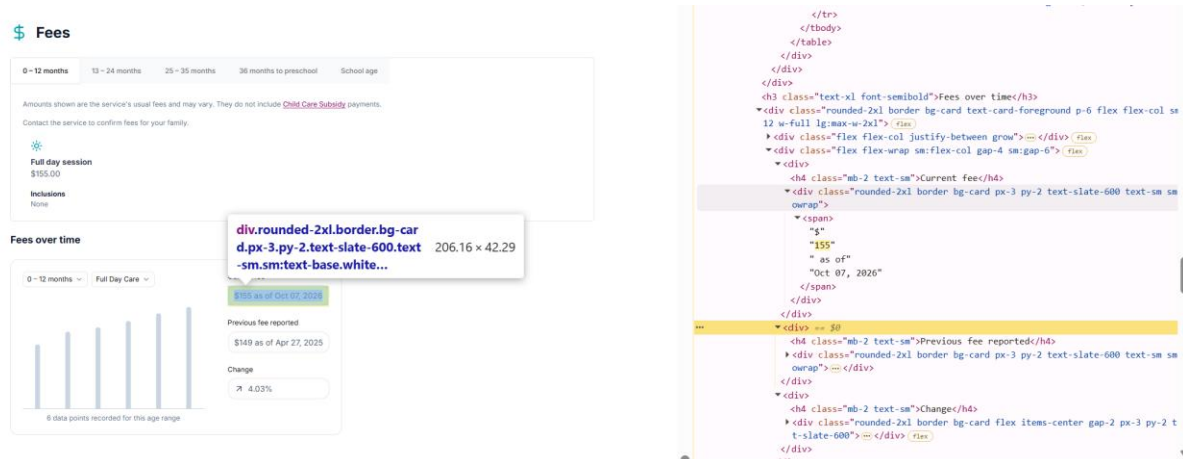


The dashboard means there are multiple numbers for me to extract from the webpage, so I needed to locate where these are stored. The age of children, and whether it was full day care / half / casual etc. are both toggleable. Finally the 6 data points are recorded over time.

Step 2: Right click on the webpage and click inspect. The following steps I worked with Claude to identify:



Step 3: Locate where the data was stored. Below is an example of where \$155 is stored. Working with Claude I located where the entire dataset is stored – embedding it as JSON in `<script id="__NEXT_DATA__">`.



Step 4: I then ask Claude to build a retrieval script in python since python is commonly used for web scraping. This part is given in phase2_historical_fees.

Author notes:

I do not mean any harm to the centres in scraping child care fees. It is also not perfect. This was done to show my research method. I also have not rerun this code since 25022026 and not sure if there have been updates to the website such that the code doesn't work anymore, but the steps broadly remain the same.

For my thesis I undertook ABS datalab training which was a half day course on handling sensitive datalab data which I've applied to this scraped sample set. None of the pricing data is explicitly provided in the data and follow these ABS guidelines.

DATA CLEANING:

I assume the price charged between missing periods is the same as the last updated price, i.e. if \$100 was charged on Feb 15 2025, and \$105 was charged on October 15 2025, then \$100 was charged everyday up until Oct 15 2025.

I transform the data into quarters and apply the above logic. This is located in the R file, `Cleaning_price_raw.R`.

Scraped data analysis

Information below may explain why not all data is uploaded to StartingBlocks and the requirements around providing up to date data:

Fees

Details:

You must notify us of your current fees for each service.
You must report the fee before any reductions or discounts are applied. We publish fee information to StartingBlocks.gov.au .

Timeframe:

within 14 days of:

- receiving CCS approval (or opening the service if that occurs at a later date)
- changing your fees
- twice a year – after the end of the financial year and calendar year, including reporting no change in fees.

How to notify:

through the Provider Entry Point or your third-party software

Task card:

[How to update business details for StartingBlocks.gov.au in the Provider Entry Point](#)

Responsibility:

persons with management or control and persons responsible at the service (for their service only)

FAL reference:

Minister's Rules s55, Item 1

Hi Dorothy

Thank you for contacting StartingBlocks.gov.au.

StartingBlocks.gov.au is brought to you by the Australian Children's Education and Care Quality Authority (ACECQA).

The family focused website has a '[Find Child Care](#)' feature that is ideal for families who are looking for convenient ways to search for services in their area.

The 'Find Child Care' feature allows parents to:

- search via keyword or postcode
- review mapped service locations
- filter by vacancies, service/session types, service ratings
- pick, save and share favourites with family and friends.

Daily fees and inclusions

The information in our 'Find Child Care' section on the Starting Blocks website is drawn from multiple sources, including the Provider Entry Point (PEP).

Services are responsible for keeping their information up to date to ensure the information displayed is correct.

If a service wants to update their service details such as vacancies, fees and inclusions, they will need to access their PRODA account. The service data they enter flows through to display on StartingBlocks.gov.au Find Child Care tool.

For further details about fees and inclusions such as hours of care and session lengths, you will need to contact the service directly.

Useful resources

StartingBlocks.gov.au also provides useful resources for families who are looking to find quality education and care services. Resources include information on [choosing a service](#) and [what to look for](#) in a quality early learning service.

StartingBlocks.gov.au wants to keep it simple and aims to assist families with fast, easy and quality focused 'Find Child Care' feature that they can use at anytime, anywhere. Visit StartingBlocks.gov.au for more helpful tips and information.

I hope this information is helpful.

Kind regards
Elliander

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Discussion: bias of data

I conduct a t-test and reject the null hypothesis that the scraped sample set is random. There is a positive bias of 44.6 cents. The t-test was constructed using information from the Department of Education below:

Re: Average Hourly Fee calculation ETKN:021000001784 [SEC=OFFICIAL]



Education - EC Data Requests <ECDataRequests@education.gov.au>

To: You

Cc: Education - EC Data Requests



Mon 30/03/2026 1:40 PM

Hi Dorothy,

Thank you for your interest in child care data.

The mean fee per hour in the CCS quarterly data tables (Table 9.1 and Table 10.1) is calculated as follows:

1. the average fee per hour is calculated for each individual service:

$$\text{fee per hour} = \frac{\text{sum of all charged fees for the service in the quarter}}{\text{sum of all charged hours for the service in the quarter}}$$

2. then for each SA3 region the mean is calculated from all the services individual fee per hours values in that SA3 region. This results in the overall mean fee per hour value for that SA3.

This analysis is for all children regardless of age. The hours used in this calculation is the charged hours of care, which is also used in the calculation of the weekly hours table, but does not utilise the average hours per week value rather considers the actual charged hours.

Penny

Data Collections, Analysis and Reporting
Australian Government Department of Education
Working on Ngunnawal Country

The scraped data has the following coverage across SA2's – it is not complete.

Scrape coverage vs NQS — year_quarter 100

